Paleomagnetic results from the Cambrian and Ordovician sediments of Bornholm (Denmark) and Southern Sweden and paleogeographical implications for Baltica

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[1] If apparent polar wander paths (APWP) cross, the question arises how to prove the older magnetization to be primary and not just a younger overprint. This problem is typically met in areas affected by percolating mineralizing fluids and/or heating due to a younger regional igneous activity. The Permian magnetic overprint is the classical example. Earlier paleomagnetic studies over the Lowermost Cambrian Nekso Sandstone Fm (NSF) of Bornholm (Denmark) yielded a characteristic remanent magnetization (ChRM) similar to the Permian directions for Baltica. Since a possible reason could be a chemical overprint, we checked whether this phenomenon did take place on a regional scale. Some samples therefore were collected from other Lower Cambrian clastics of Bornholm and Southern Scandinavia. In result we show that the well-grouped and stable ChRM of the NSF contrasts with fairly chaotic, soft, and badly preserved magnetizations of the Balka, Hardeberga, Mickwitzia, and Lingulid sandstones of Bornholm and Southern Sweden, thus not indicating widespread paleomagnetic overprint. We demonstrate that the ChRM of the NSF is most probably of syndepositional/early diagenetic origin and its similarity to the Permian direction for Baltica is only casual. We propose a normal polarity and a near-equatorial position on the Southern Hemisphere for Baltica in the early Cambrian time, as well as a more complicated trend of the APWP for this paleocontinent than envisaged by other authors. Paleomagnetic results from the Arenigian limestones of the Laesaa Formation (Bornholm) that yield excellently defined but most probably only secondary components are also presented. *INDEX TERMS:* 1519 Geomagnetism and Paleomagnetism: Magnetic mineralogy and petrology; 1527 Geomagnetism and Paleomagnetism: Paleomagnetism applied to geologic processes; 1533 Geomagnetism and Paleomagnetism: Remagnetization; 8157 Tectonophysics: Plate motions—past (3040); 9614 Information Related to Geologic Time: Paleozoic; *KEYWORDS:* paleomagnetism, Cambrian, paleogeography, Baltica, Bornholm, Scania

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1. Introduction

[2] The drift history of continental fragments after the break-up of the late Precambrian super-continent Rodinia and their reassembly into the late Paleozoic megacontinent Pangea is actually a matter of vital debate [*Dalziel*, 1992, 1997; *Torsvik et al.*, 1996; *Kirschvink et al.*, 1997; *Torsvik et al.*, 1998; *Evans et al.*, 1998; *Torsvik and Rehnström*, 2001]. However, many elements in reconstruction of the Late Precambrian-early Phanerozoic paleogeographic scenarios are hypothetical. This is partly due to the still few paleomagnetic data covering long timespans, the lack of longitudinal constrains due to the rotational symmetry of the global dipole field, and possibly also a nondipolar behavior for parts of the Paleozoic and older times still to be resolved [*Pesonen and Halls*, 1983; *Van der Voo and Torsvik*, 2001].

[3] Paleogeographic reconstructions for the Neoproterozoic and Cambrian thus are still in lack of reliable paleomagnetic data [*Torsvik et al.*, 1996; *Smethurst et al.*, 1998], though new paleopoles for 535 and 500 Ma recently have been reported [*Torsvik and Rehnström*, 2001]. Paleomagnetic poles obtained from Cambrian successions over the world often suffer from a low or unknown reliability and a significant scattering, even within a single crustal plate [*Van der Voo*, 1990, 1993; *Symons and Chiasson*, 1991; *McKerrow et al.*, 1992; *Torsvik et al.*, 1992, 1996]. This results in incoherence of the paleogeographic recon-

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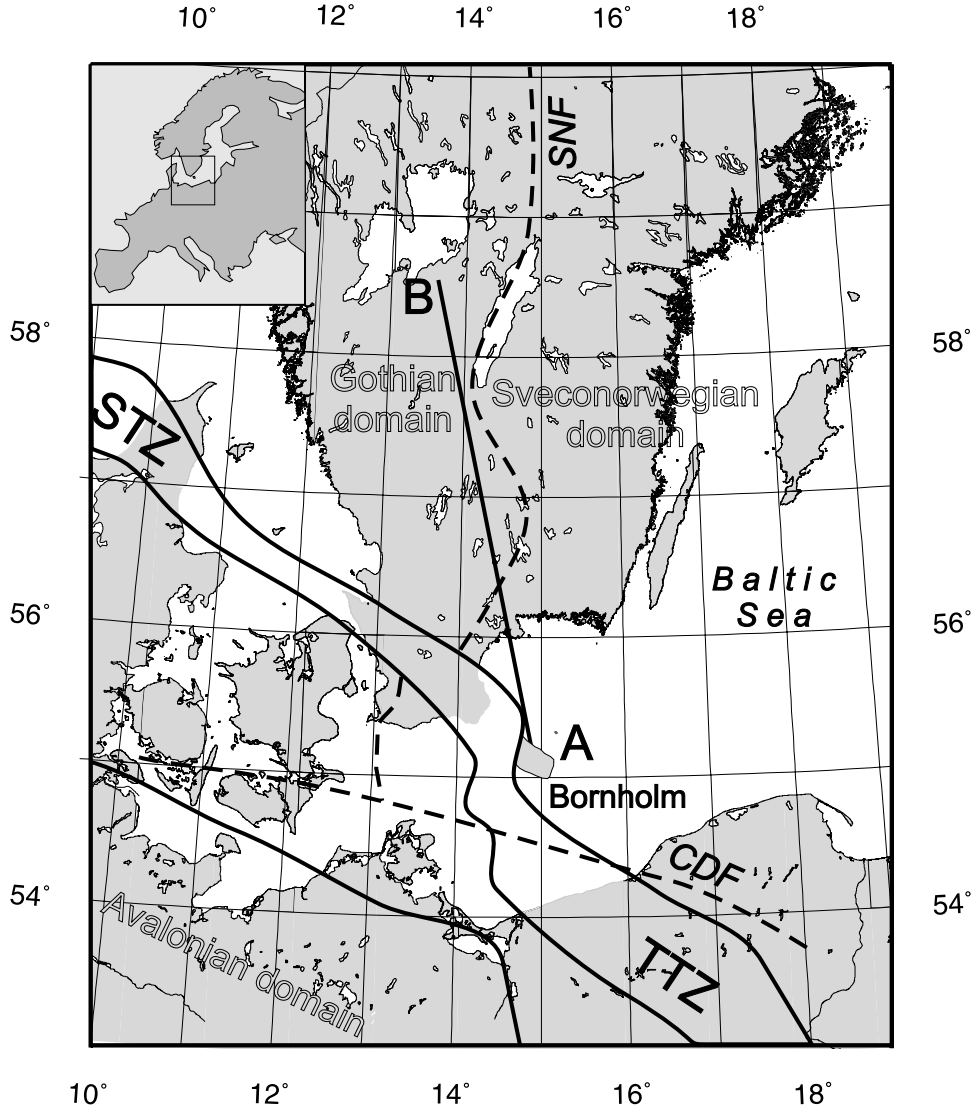


Figure 1. A position of Bornholm with respect to the main structural units of the Baltic sector of the East European Craton. Insert shows the general situation of the study area on the map of Europe. TTZ is Tornquist-Teisseyre Zone; CDF is Caledonian Deformation Front; SNF is Sveconorwegian Front. Profile A-B is shown in Figure 3.

structions. Depicted configurations [e.g., *Dalziel*, 1992; *McKerrow et al.*, 1992] differ significantly in latitudinal positioning of the major crustal blocks. The current early Cambrian paleomagnetic pole position for North America indicates high paleolatitude for Laurentia during this time [*Symons and Chiasson*, 1991] and fits the reconstruction by *Dalziel* [1992] but contrasts with an equatorial positioning of Laurentia as presented by *McKerrow et al.* [1992]. On the other hand, McKerrow et al.'s configuration situates Scandinavia in subtropical latitudes, while *Dalziel* [1992] places it at a latitude of 60°S.

[4] The Early Cambrian paleogeography is particularly debated since paleomagnetic data collected from Siberia, Baltica, Laurentia, and Gondwana may be interpreted in terms of fast (535–517 Ma) rotation and substantial latitudinal drift of these paleocontinents [*Kirschvink et al.*, 1997] (for discussion see *Torsvik et al.* [1998] and *Evans et al.* [1998]). Questions upon the reality of the Cambrian loop for Baltica's apparent polar wander path (APWP) have recently been raised by *Smethurst et al.* [1998] and *Torsvik and Rehnström* [2001].

[5] Any new paleomagnetic data from Cambrian rocks may therefore significantly contribute to this discussion. In this context we present results from the Danish island of Bornholm, situated in the southern Baltic Sea and from southernmost Sweden (Figure 1).

[6] On Bornholm the Phanerozoic sedimentary succession starts with the Lower Cambrian Nekso Sandstone Fm (NSF), which was paleomagnetically investigated earlier by *Prasad and Sharma* [1978, 1980] and by us [*Abrahamsen and Lewandowski*, 1994; *Lewandowski and Abrahamsen*, 1998]. In our previous contributions we contested the conclusion by *Prasad and Sharma* [1978] of a primary origin of the characteristic magnetization of the Nekso sandstones and attributed it to a Permian remagnetization. Taking into account the striking similarity of the ChRM of

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the NSF to the well-known reversed polarity Permian direction for Baltica, the hypothesis of a secondary origin of this magnetization seemed quite plausible.

[7] We tested this hypothesis procedingly, however, by further extensive sampling of other sites of the Lower Cambrian formations on Bornholm (NSF, Balka Sandstone, Broens Odde siltstone) [see *Lewandowski and Abrahamsen*, 1999] and in Southern Sweden as well (Hardeberga, Mickwitzia, and Lingula Sandstones) [see *Lewandowski et al.*, 1999]. The intention was to check whether a paleomagnetic overprint could have been caused by secondary magnetic phases originating from fluids that possibly had traveled laterally along the contact between the basement and the Lower Cambrian sediments. Some samples therefore were collected from beds which, just like the NSF, are resting directly on top of the weathered old basement surface (the Mickwitzia Sandstone in the Hällekis abandoned mine, Västergötland, S. Sweden). Should the assumption of a Permian remagnetization be correct, we would expect to see the same remagnetization effect in other Lower Cambrian rocks, overlying the crystalline basement of the southern Scandinavia.

[8] In the following we present the results of this regional test, followed by a comprehensive reanalysis of our Nekso data, including new results obtained after additional samplings. Additionally, we also present new paleomagnetic data from the Lower Ordovician (Arenigian) limestones of the Læseaa Formation.

2. Geology

[9] Structurally, the island of Bornholm (55°N/15°E) is a horst (Figure 1) situated at a kink of two converging fault zones, namely the Sorgenfrei-Tornquist Zone (STZ) [see *EUGENO-S*, 1988] and the Tornquist-Teisseyre Zone (TTZ) [*Pegrum*, 1984; *Guterch et al.*, 1986; *Liboriussen et al.*, 1987; *Dadlez*, 2000]. At present, TTZ is considered a heterochronic feature, probably of pre-Silurian foundation [*Pożaryski et al.*, 1982; *Berthelsen*, 1992, 1993, 1998]. The Bornholm block has remained an integral part of Baltica at least since c. 1360 Ma [*Abrahamsen and Lewandowski*, 1995], and hence we expect to encounter the same APWP for both units since after.

[10] Evidenced from cross-bedding and planar stratification, NSF was deposited in a fluvial and aeolian environment during early Cambrian time, resting directly upon the exposed Precambrian basement [*Poulsen*, 1978; *Clemmensen and Dam*, 1993]. Mineralogically, it is of an arkosic to subarcosic feldspar-rich composition [*Møller and Friis*, 1999]. The almost horizontally lying, well bedded, and finegrained to silty sandstones gradually gives way to the shallow marine Hardeberga (Balka) Sandstone Formation (HSF). Both sandstone formations are highly compacted and thoroughly cemented by quartz during the Late Silurian/Early Devonian burial, and consequently have very little porosity at present (1% on average) [see *Møller and Friis*, 1999]. NSF and HSF are overlaid by silt- and fine sandstones of the Laesaa Formation [*Surlyk*, 1980; *Pedersen*, 1989], both being ascribed to the Lower Cambrian *Schmidtiellus mickwitzi* Zone and *Holmia kjerulfi* Assemblage Zone [*Moczydłowska and Vidal*, 1992]. This is in keeping with the earlier conclusions by *Poulsen* [1978] who, on the basis of a lithostratigraphic correlation, pointed to an early Cambrian age (Lower Tommotian) of the NSF. According to the current isotope chronology [see *Tucker and McKerrow*, 1995] it corresponds to an age of ca. 545 Ma. Toward the south, the NSF deepens under younger deposits and it is known from boreholes in the Polish part of the East European Platform (EEP) as a member of the Vendian/Cambrian Żarnowiecka Formation [*Lendzion*, 1983].

[11] Paleocurrent evidence [*Bruun-Petersen*, 1971] indicates that streams flowing from NNW deposited the lower part of the NSF. Hence a considerable part of the sediment was derived from S. Sweden [*Jensen*, 1977]. The variegating, in some places intensively red colors of intercalations within the otherwise grayish to yellowish NSF suggest a generally warm climate during the deposition [cf. *Witzke*, 1990], providing the red color to be of genuine origin [e.g., *Pittman*, 1979]. *Poulsen* [1978], *Surlyk* [1980], *Dam and Clemmensen* [1988], *Berthelsen* [1989], *Pedersen* [1989], *Moczydłowska and Vidal* [1992], and *Møller and Friis* [1999] have presented comprehensive studies on lithology, lithostratigraphy, and paleoenviroment of the NSF and overlying successions.

[12] Studies on the stable isotope composition [*Buchardt and Nielsen*, 1985; *Buchardt et al.*, 1997] as well as on the state of preservation of microfossils [*Moczydłowska and Vidal*, 1992] point to thermal maturation of the overlain successions (and implicitly also of the NSF) in the range of 90°–150° C, probably caused by a 2–3 km [*Buchardt and Nielsen*, 1985; *Buchardt et al.*, 1997] or even 4 km [*Vejbæk et al.*, 1994] later burial during Late Silurian to Early Devonian time. The Læseaa Formation consists of thin-bedded Arenigian limestones, which are exposed and accessible in the southermost part of Bornholm (Figure 2) (for a more detailed description, see *Surlyk* [1980]).

3. Paleomagnetic Sampling, Laboratory Techniques, and Analysis

[13] The Cambrian collection from Bornholm comprised 185 samples. The major sampling of the NSF was made in two quarries at Gadeby (northern and southern outcrops) and at Slamrebjerg (1 km NE of Gadeby), both situated just north of the road between Nekso and Rønne (Figure 2). At Gadeby and Slamrebjerg quarries a total of 132 independently oriented samples (124 samples cored perpendicularly to bedding plane and eight hand samples) were collected from 34 sites, spacing more or less uniformly along the 23 m thick, horizontally laid sequence of the NSF. (A site is defined here as a separate place within the outcrop, confined to 1–2 beds).

[14] This collection was later supplemented by 20 hand samples taken from exposures at the coast between Nekso and Balka (Lange Skanse) from the NSF. Sampling of the slightly younger Balka Quartzite was made in the old quarry of Smaalyngovaskut (15 hand samples) nearby Pedersker, while the Green Shales Formation (Grønne Skifre, GSF) was sampled in the vicinity of Skelbro (18 hand samples). Furthermore, hand samples from slightly younger formations (see Figure 3) were collected from the Hardeberga, Mickwitzia, and Lingulla Sandstones in Scania, S. Sweden (a total of 49 hand samples). Orientations were made

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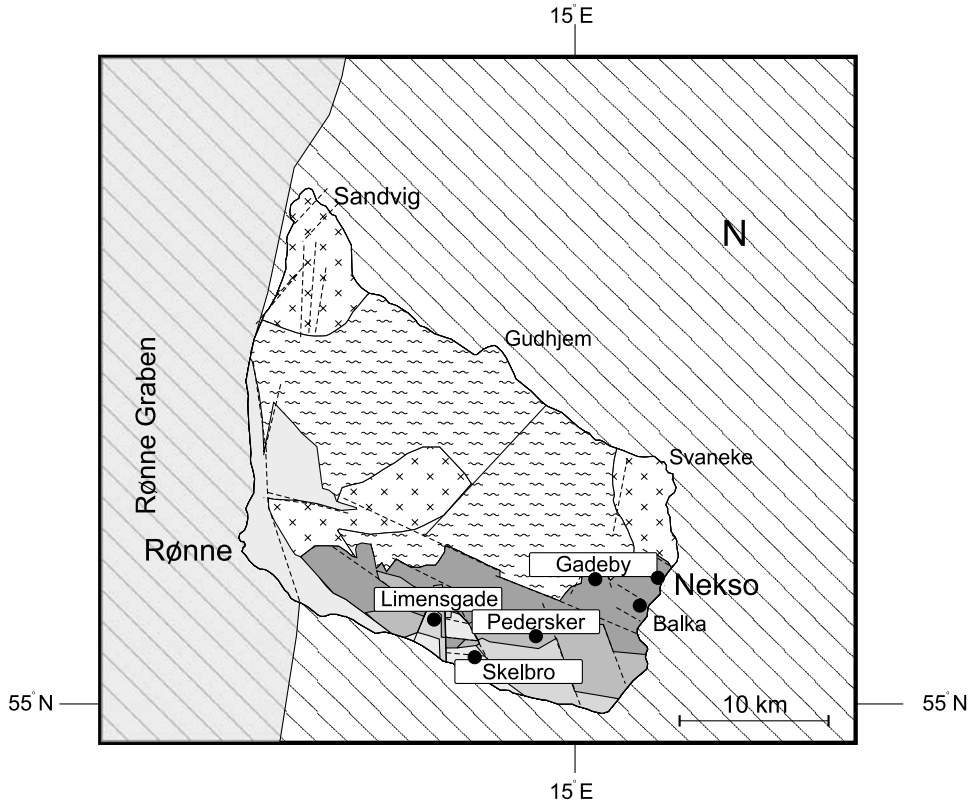


Figure 2. Geological map of Bornholm (modified after *Abrahamsen and Lewandowski* [1995] and *Berthelsen* [1989]); sampling localities are shown by dots (further information in the text). Oblique lines are the basement covered by the Baltic Sea; crosses/wavy open signature are the Precambrian basement at the surface (granites and gneisses, respectively); dotted hatching is the post-Triassic sediments in the Rønne graben west of Bornholm; deep gray is the Cambrian deposits; medium gray is the Ordovician limestones; light gray is the post-Paleozoic deposits.

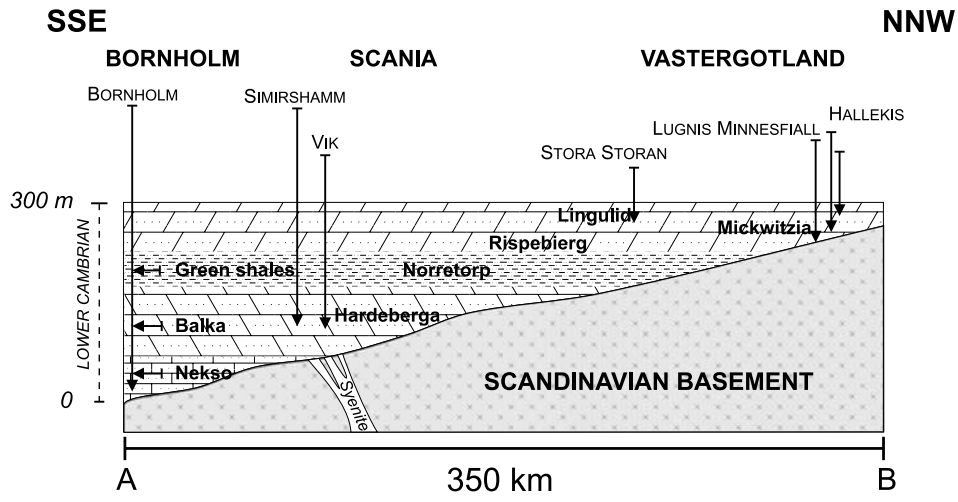


Figure 3. Simplified cross-section from Bornholm through South Sweden (see line A-B, Figure 1) at the end of the Cambrian, based on the geological bedrock map of Scania (SGU 1988). The prograding deposition of Cambro-Ordovician sediments from S to N due to the basement topography along the northern shores of the Iapetus Ocean/Tornquist Sea is illustrated. Erosion of the syenites in Sweden and fluvial/shallow water transport about 100 km to the S could supply Ti-hematites during early Cambrian time to the Nekso area, but this source was soon after covered by the Balka Quartzite and younger deposits due to basin deepening. Sampling places are shown against a regional lithostratigraphic division.

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essentially by means of a sun compass when cores were taken, while for orientation of hand samples a magnetic compass was used. The total collection of the Cambrian sediments from Bornholm and Southern Sweden comprised 234 oriented samples.

[15] An average bedding dip of 2°S for the Cambrian rocks has been considered negligible for applying a tectonic correction. The Arenigian Orthoceras limestones were sampled from horizontally to near-horizontally lying beds at Limensgade (eight hand samples) and in a small quarry nearby Skelbro (seven hand samples).

[16] Identification of the magnetic minerals was done by thermomagnetic analysis, using tiny cylindrical cores (8 mm × 8 mm). The specimen was first subjected to magnetization in a pulse magnetizer to produce an induced remanent magnetization (IRM), and after this it was measured in a spinner magnetometer while continuously heated in zero field in a furnace combined with the magnetometer (for apparatus description, see *Kadzialko-Hofmokl and Kruczyk* [1976]). During the experiment the unblocking temperature (T_{ub}) spectrum was plotted against the decaying IRM. A Vibrating Sample Magnetometer (VSM) by Molyneux was used to investigate the elements of the hysteresis loops.

[17] Thermal demagnetization was made by using a Magnetic Minerals MM-1 furnace. Each specimen was heated stepwise up to as much as 650°C and cooled in a zero magnetic field. After each demagnetization step, the remaining magnetization was measured either on a 2G SQUID cryogenic magnetometer with a residual internal field below 3 nT and a noise level of about 5 μA/m or with a JR-4 spinner magnetometer. Both the furnace and the SQUID magnetometer were operated inside Helmholz coils, reducing the ambient geomagnetic field by 95%. The experiment continued until the remanent intensity dropped down to the magnetometer noise level unless a spurious magnetization appeared in the specimen. Measurements of magnetic susceptibility and its anisotropy (AMS) were performed by means of a Czech KLY-2 susceptibility bridge.

[18] Orthogonal projections of the remanence vector endpoints during stepwise demagnetizations were plotted for every specimen. Paleomagnetic Data Analysis (PDA) software by M. Lewandowski et al. (unpublished manuscript, 1997), employing principal component analysis [*Kirschvink*, 1980], was used to calculate the characteristic NRM components (ChRM) from the demagnetization data and to plot the demagnetization diagrams. Great circle paths, defined by the remanent magnetization (RM) vectors during the stepwise cleaning procedure, were also applied to determine the best-fitting planes with angular standard deviations less than 5°.

[19] As we were dealing with quite scattered paleomagnetic directions, we applied a Gaussian method to count the point density and to determine statistically significant populations. *Jowett and Robin* [1988] have shown that to a first approximation a peak count greater than (E + 4S) is higher than expected for a uniform population and is considered significant at the 95% confidence level. Here E is the expected count that should arise if the data set was uniformly distributed, while S (Sigma) is the standard deviation of the counts variation around the expected value for a uniform distribution. The counting was performed utilizing the *SpheriStat* program [*Stesky*, 1995] and using an extra fine grid with the E = 3∗Sigma method, which corresponds to Kamb's fixed-circle method [see *Stesky*, 1995]. The same program was used for presentation of AMS directions.

[20] Standard *Fisher* [1953] statistics were used to calculate the characteristic mean direction for a statistically significant population. A corresponding paleopole position was calculated and plotted relative to the APWP using the GMAP software by *Torsvik and Smethurst* [1997].

4. Paleomagnetic Results

4.1. Cambrian Rocks: Nekso Sandstone Formation (NSF), Bornholm

4.1.1. Magnetic Carriers

[21] Remanence carriers thus identified are (1) hematite in the forms of detrital grains (specularite), pseudomorphoses after magnetite (martite), and pigment coatings and (2) ilmenite-hematite exsolution detrital grains of Ti-poor hematites. This mineral assemblage of the opaque minerals has been proved by microscopic studies [*Jensen*, 1977] and is in line with results presented by *Prasad and Sharma* [1978] and *Clemmensen and Dam* [1993].

[22] Low-Ti hematite is the dominating magnetic carrier in the red stuff of the NSF, and its presence is confirmed by the thermomagnetic analysis as a phase with maximum T_{ub} (unblocking temperature) around 640°C (Figure 4a). Microscopic observations point to a relatively early development of hematite coatings over the detrital quartz grains (H. Friis, personal communication, 2000) so that the hematite pigment could acquire a true crystalline growth chemical remanent magnetization (CRM) pene-contemporaneous with the sedimentation.

[23] Microscopic studies [*Prasad and Sharma*, 1978] proved a presence of Ti-poor hematite grains. According to *Jensen* [1977], the grains of ilmenite-hematite exsolution intergrowths are generally fresh, but occasionally original ilmenite grains have been strongly attacked by alterations of the so-called beach sand-type. In the thermomagnetic analyses, presence of low-Ti hematites may be inferred from a maximum T_{ub} about 640°C, combined with very low intensities of induced remanent magnetization (Irm). Coercivities of these specimens vary within one order of magnitude and they are inversely proportional to the normalized remanence values (Figure 5). Some specimens, however, show higher values of Irm with the same T_{ub} range, what may be indicative for the presence of maghemite (Curie temperature of 645°C) [*Dunlop and Özdemir*, 1997]. Coming from neighboring magmatic rocks, these specular low-Ti hematite and maghemite grains are potent candidates for carrying detrital (DRM) and postdetrital (pDRM) remanent magnetization [e.g., *Steiner*, 1983].

[24] The thermomagnetic analysis did not show any sign of magnetic sulfide (phyrrotite), although minor amounts of sulfides have been described elsewhere in relation to the Nekso Sandstone [*Bøgvad*, 1940; *Bøggild*, 1943]. Contrary to *Prasad and Sharma* [1978], we did recognize predom-

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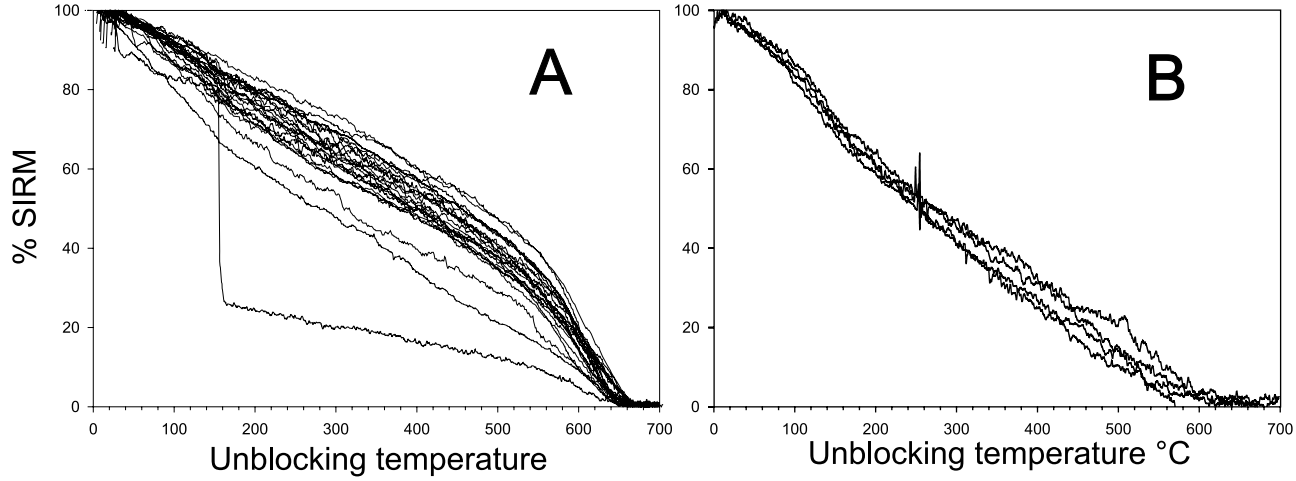


Figure 4. Thermomagnetic curves, depicting intensities of saturation magnetization (SIRM, arbitrary units) against temperature. (a) Examples from NSF. Maximum unblocking temperatures around 645°C are indicative of low-Ti hematite [*Dunlop and Ozdemir*, 1997]. One curve with a rapid drop in the intensity points to a presence of goethite. (b) Examples from the Arenigian limestone of Skelbro. Maximum T_{ub} around 580°C is indicative for magnetite [*Dunlop and Ozdemir*, 1997].

inant orientations of anisotropic magnetic susceptibility (AMS) in 60 specimens measured. The minimum axes of the AMS are oriented essentially downward, while the maximum axes align along the NW-SE direction. Both features (see Figure 6) are characteristic for the sedimentary type of an AMS distribution [*Tarling and Hrouda*, 1993], developed in a NE-SW current-flow environment, in contrast to the NNW-SSE transport direction inferred by *Bruun-Petersen* [1971]. Such AMS pattern is speaking in favor of a preservation of an original sedimentary-magnetic texture of the NSF.

4.1.2. Principal Component Analysis (PCA)

[25] The NRM of the NSF show a great variability of composition between the specimens even within the same site. Differences in the component's orientation, spectral structure, and/or polarities result in dissimilarities of individual demagnetization patterns and hamper the paleomagnetic interpretation. The examples shown in Figure 7 are characteristic for this variability.

[26] Only ca. 40% of the collection yielded interpretable results. The PCA revealed a variety of poorly clustered NRM components with a dominating group of single polarity directions pointing SSW with a shallow negative inclination. Taking into account the identified assembly of magnetic carriers and the frequency of T_{ub} observed for the whole population in the course of thermal demagnetization, three groups of components can be distinguished (Figure 8): (1) low temperature components A (removed up to 400°C), (2) intermediate components B for the temperature range 400°C–625°C, and (3) hard components C, featuring T_{ub} above 625°C.

[27] In the temperature above 625°C the demagnetization paths display zig-zag trends; thus linear segments defined by PCA may reflect a remanence created during the demagnetization experiments above 625°C. We also cannot exclude that some of the PCA lines fitted to three or four points may be casual even for the intermediate T_{ub} range, taking into account the uneven trends of many of the demagnetizon curves.

[28] The ChRMs show statistically significant concentrations toward SW with shallow negative inclinations (Figure 8). In order to ensure the highest reliability for determination of a characteristic mean, only those PCA lines fitted to more than four points and with ASD values <6° were taken into account to calculate the mean direction. Calculation has been performed for n = N = 16 components (12 of B-type and 4 of C-type), selected from the whole population of components fulfilling the above criteria, using statistical methods of *Jowett and Robin* [1988]. The directional coordinate obtained is (D = 194°, I = −12°), with a precision parameter of k = 32 and the oval of 95% confidence level α_{95} = 7°. The corresponding paleopole, assumed to be southern, has the coordinates 177°E/40°N.

4.1.3. Great-Circle Method

[29] If the blocking temperature spectra of two (or more) NRM components overlap, the PCA is not efficient in the determination of their directions. Since this was a typical case for our NSF specimens, we have applied the great-circle method of *Halls* [1976] to discriminate the recorded components. For flat-lying strata as in the case of the NSF, this method will be useful providing that intensity, orientation, and polarity as well as a spectral composition of all components will be sufficiently differentiated to introduce an appreciable scatter of the great-circle planes. If this condition is fulfilled, a point of great-circles convergence can be resolved with sufficient precision, giving an orientation of the paleomagnetic field direction, in which a given component has been acquired.

[30] During the thermal cleaning of the NSF specimens we thus observed a well-defined directional migration of the remanence vectors along great-circle planes towards their point of convergence (Figure 9a). Groups of specimens that show similar components of the NRM directions will show crossing points projected into more or less restricted areas on the stereonet (Figure 9b).

[31] The density distribution analysis of all great-circles crossing points (n = 7711, see Figure 9c) enabled us to

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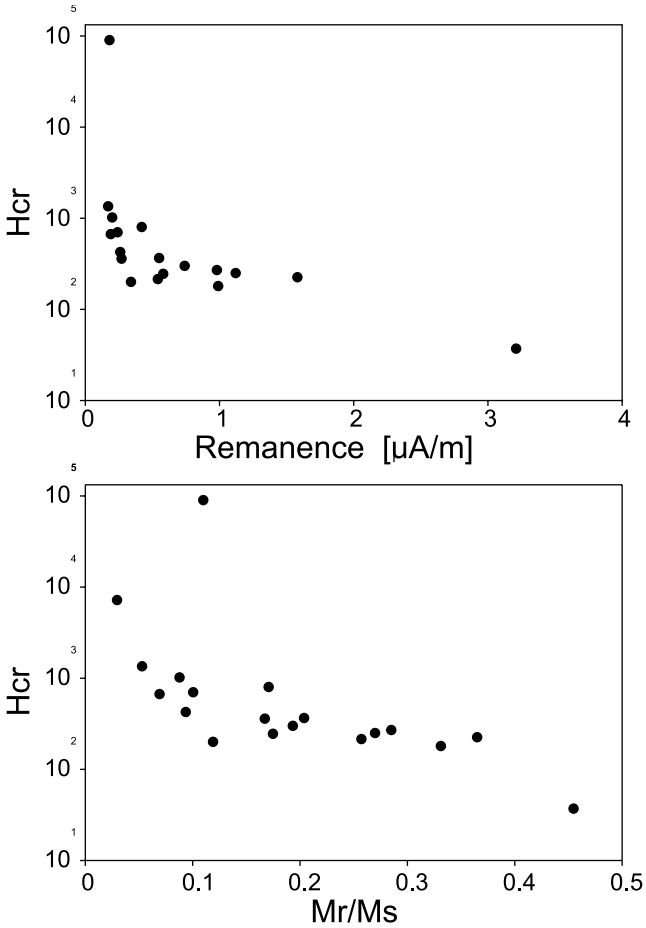


Figure 5. Hysteresis parameters for 18 specimens of the NSF, measured with a Vibrating Sample Magnetometer (VSM, Molyneux type). Average values of the coercive force (Hcr) are characteristic for hematite, remanence values <2 may indicate a presence of fine grained fraction [*Dunlop and Özdemir*, 1997, and references therein]. The smeared distribution of the Mr/Ms ratio values may be due to the grain size variation.

resolve orientations of the preferred paleomagnetic axes to which the great circles converge or from which they diverge (depending on the relative magnetic hardness of the components involved). The density distribution of all crossing points is shown in Figure 9d (only the lower hemisphere is presented, as the upper hemisphere shows an antipodal image). Clusters labeled as A_C, B_C, C_C, and D_C are all statistically meaningful, peaking around the paleogeomagnetic axes. Cluster A_C corresponds to the ChRM obtained by the PCA. The remaining axes B_C, C_C, and D_C are poorly defined by the PCA; nonetheless they may represent Silurian records acquired during burial and uplift of the NSF, since the B_C, C_C, and D_C orientations are fairly similar to the Silurian/early Devonian paleofield directions of Baltica.

4.2. Other Lower Cambrian Sediments of Bornholm and Scania (S. Sweden)

[32] We present the results of our paleomagnetic investigations of the Lower Cambrian sediments, which are resting atop of the NSF (Figure 3), in a collective manner only because they have yielded very disappointing, scattered results. The Balka sandstones were investigated at the Pedersker area (Bornholm), the GSF was investigated at Broens Odde (Bornholm), the Hardeberga sandstones were investigated at Simirsham and Vik (Southern Scania), the Mickwitzia sandstones were investigated in the Hällekis and Lugnis abandoned mine, and the Lingulid sandstones were investigated in the Lidkøping area, northern Scania (Figures 2 and 3).

[33] Saw-toothed, mostly not interpretable demagnetization diagrams (Figure 10) were in common for almost 80 hand samples, collected along the profile from the island of Bornholm up to the shore of the great Lake Vänern in South Sweden (see Figures 1 and 3 for sampling localities). Only a few out of a dozen thermomagnetic curves have allowed for a determination of the magnetic carriers, which could be identified as magnetite (Mickwitzia and Balka) and hematite. Zig-zag trends in the demagnetization curves probably resulted from the presence of multidomain

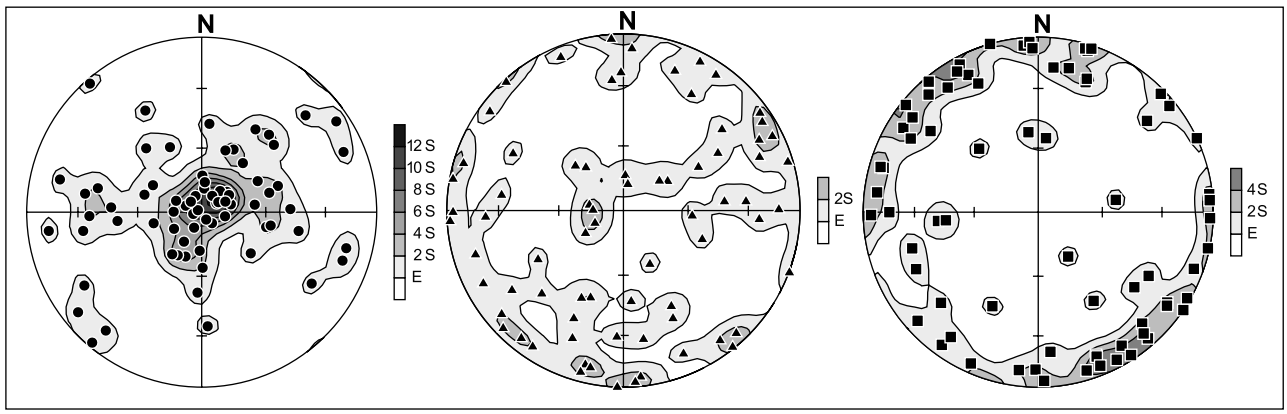


Figure 6. AMS (anisotropy of the magnetic susceptibility) in the NSF. Orientation of minimum (dots), intermediate (triangles), and maximum (squares) axes of the AMS ellipsoids are shown. The pattern distribution is characteristic for sedimentary conditions, with the minimum axes oriented perpendicular to the sedimentation plane. Note the concentration of maximum axes along the NW-SE, indicative of NW-SE currents flow. Contours of density are scaled in standard deviation values (S), relative to the value of E (see text for explanations). Schmidt projection.

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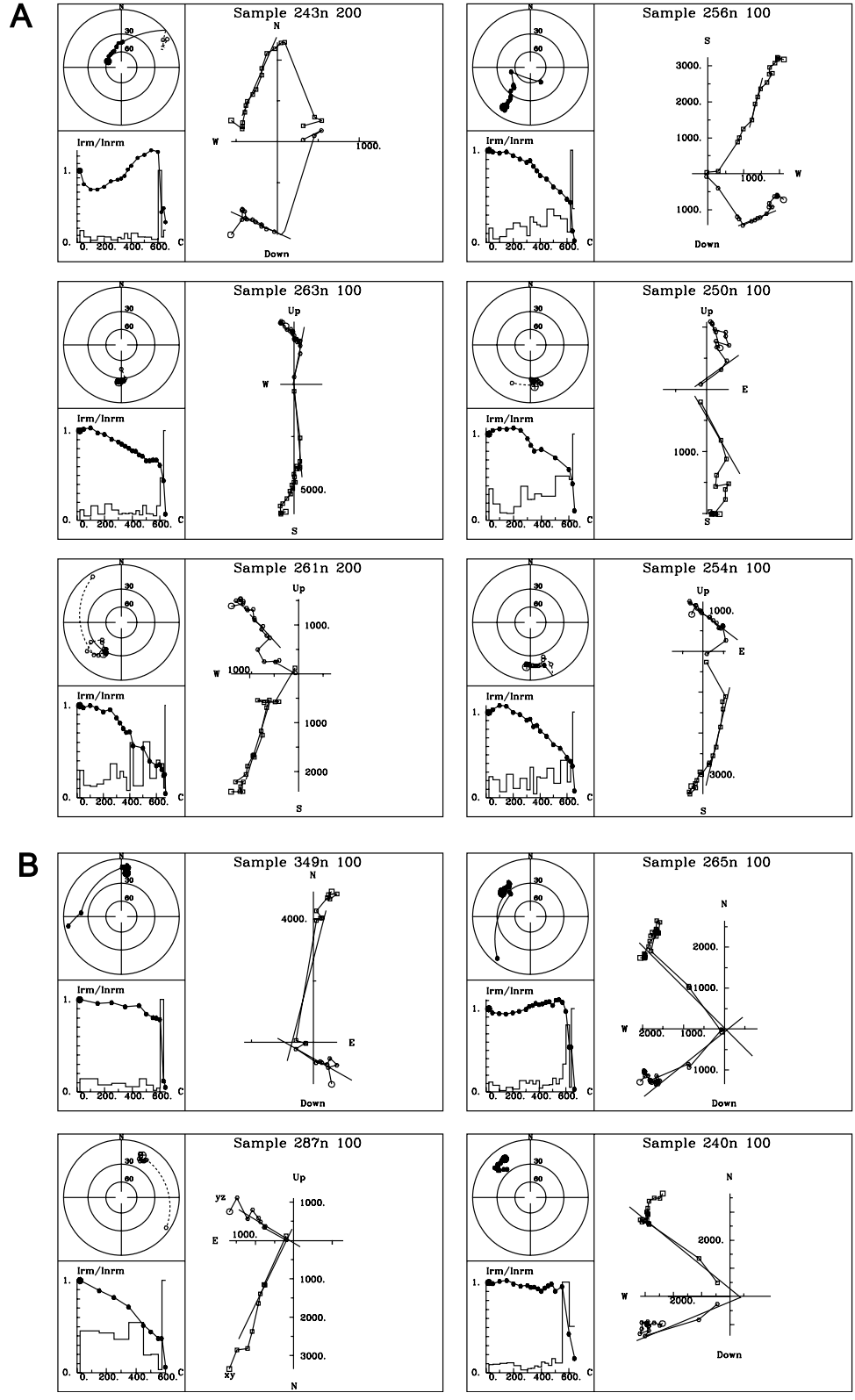


Figure 7. Typical examples of response of the NSF specimens to a thermal demagnetization. (a) Definition of the characteristic remanent magnetization (ChRM) for the NSF by means of principal component analysis; directions fitted to less than five points are shown by straight lines. (b) Examples of opposite polarity directions (not taken into account for calculation of the characteristic mean). Open/full squares are upward/downward pointing vectors. Inrm is the intensity of NRM; Irm is the intensity of remanence at the given temperature level; units in μA/m. Lines defined by PCA are shown.

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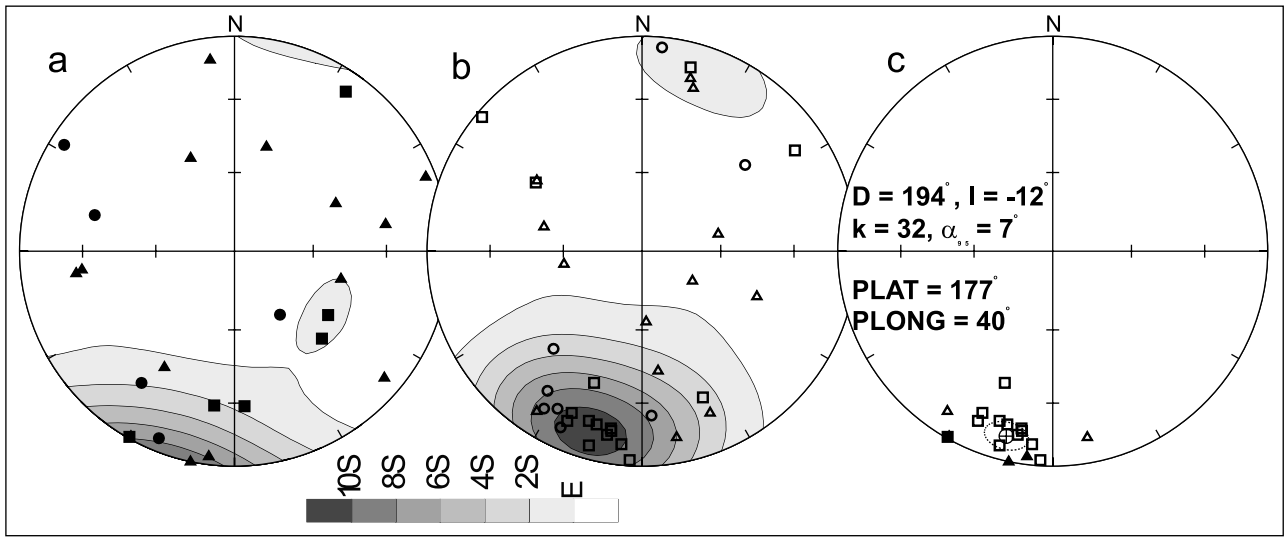


Figure 8. Determination of a characteristic mean from PCA lines obtained for the NSF. (a and b) Density distribution of lines fitted to more than four points. Contours of density scaled in standard deviation values (S), relative to the mean value (E), a limit of statistical significance is 4S [see *Stesky*, 1995]. Figures 8a and 8b show the lower and upper hemisphere, respectively. (c) Calculation of the characteristic mean from the distribution shown in Figures 8a and 8b. Dots are lines for Tub < 325°C, squares are 325°C < Tub < 625°C, triangles are Tub > 625°C. Schmidt projection.

magnetite only. No characteristic directions therefore could be identified in these rocks.

4.3. Læseaa Formation, Komstad Kalk, Limensgade, and Skelbro Localities, Bornholm

[34] Thermomagnetic analysis performed in both localities proved that magnetite is the only magnetic carriers (Figure 4b; note the maximum T_{ub} close to 580°C), in agreement with the conclusion by *Torsvik and Rehnström* [2003] for the Komstad limestones in Skelbro. Notably, neither in the work of *Torsvik and Rehnström* [2003] nor in our study was hematite identified. During A.F. cleaning, three components of NRM could be determined according to their coercivity (Figure 11). Except for the soft component, reversed polarity prevails, but one hand sample yielded high coercivity normal polarity component. The coercivity spectra partly overlap. In Limensgade, soft components are oriented similar to the present-day field direction (Figure 12a), while intermediate coercivity components (IBV) display a SSW orientation with horizontal inclination (D = 211°, I = 2°, k = 90, α_{95} = 6°, N = 7, paleopole L_{IBV} 28°S/339°E, see Figure 13). High coercivity components (HBV) point southerly with moderately steep inclination (D = 157°, I = 62°, k = 48, α_{95} = 6°, N = 7; paleopole L_{HBV} 11°N/32°E, see Figure 13).

[35] In the Skelbro the soft component is quite similar to that of Limensgade (Figure 12B), but IBV has a significantly different mean direction (D = 205°, I = −17°, k = 45, α_{95} = 5°, N = 7; paleopole S_{IBV} 40°S/343°E, see Figure 13). This is roughly similar to the mean for the intermediate blocking temperature component (IB), obtained by *Torsvik and Rehnström* [2003]. We note, however, that our S_{IBV} paleopole plots in the vicinity of the late Carboniferous paleopoles for Baltica (ca. 300 Ma), implying some 40 Ma older age for this component than that suggested by *Torsvik and Rehnström* [2003] for their IB component (possible causes are discussed later in this paper). HBV components in Skelbro show a strong, nonfisherian distribution, suggesting insufficient separation of NRM components, and making the calculation of their fisherian mean unjustifiable. A characteristic hard component for limestones in this locality, however, has been effectively identified by *Torsvik and Rehnström* [2003] by means of thermal cleaning and it was considered primary.

5. Discussion of the Magnetic Reliability

5.1. Cambrian Results

[36] In our regional-scale test on the presence or absence of a possible Permian remagnetization of the Cambrian rocks, undertaken by sampling in the Southern Sweden (see Figure 3), we found only traces of magnetic minerals (magnetite, hematite, and pyrrothite) of yet undetermined origin, carrying a lousy, weak, and unstable magnetization. No Permian overprints were found in the Mickwitzia sandstones, despite the fact that sampling was made in the beds deposited directly on the crystalline basement in the abandoned mine of Lugnis Minnesfiall (Figure 3). We therefore conclude that the rocks generally must have been impermeable for meteoric fluids at the time of extensive Permian weathering so that any stable chemical magnetic overprint was unable to develop. This is compatible with outcomes by *Buchardt et al.* [1997] and *Møller and Friis* [1999] that the main diagenetic event in the Cambrian rocks on Bornholm, including the compaction and cementation of the sandstone studied, occurred during the deepest burial in late Silurian/early Devonian time. We may note that poorly compacted quartzose and feldspathic shallow marine sediments are reported to have acquired a secondary remanence much later after their deposition [e.g., *Salem et al.*, 1998].

[37] As demonstrated above, the NSF carries a characteristic magnetization recorded in a variety of magnetic

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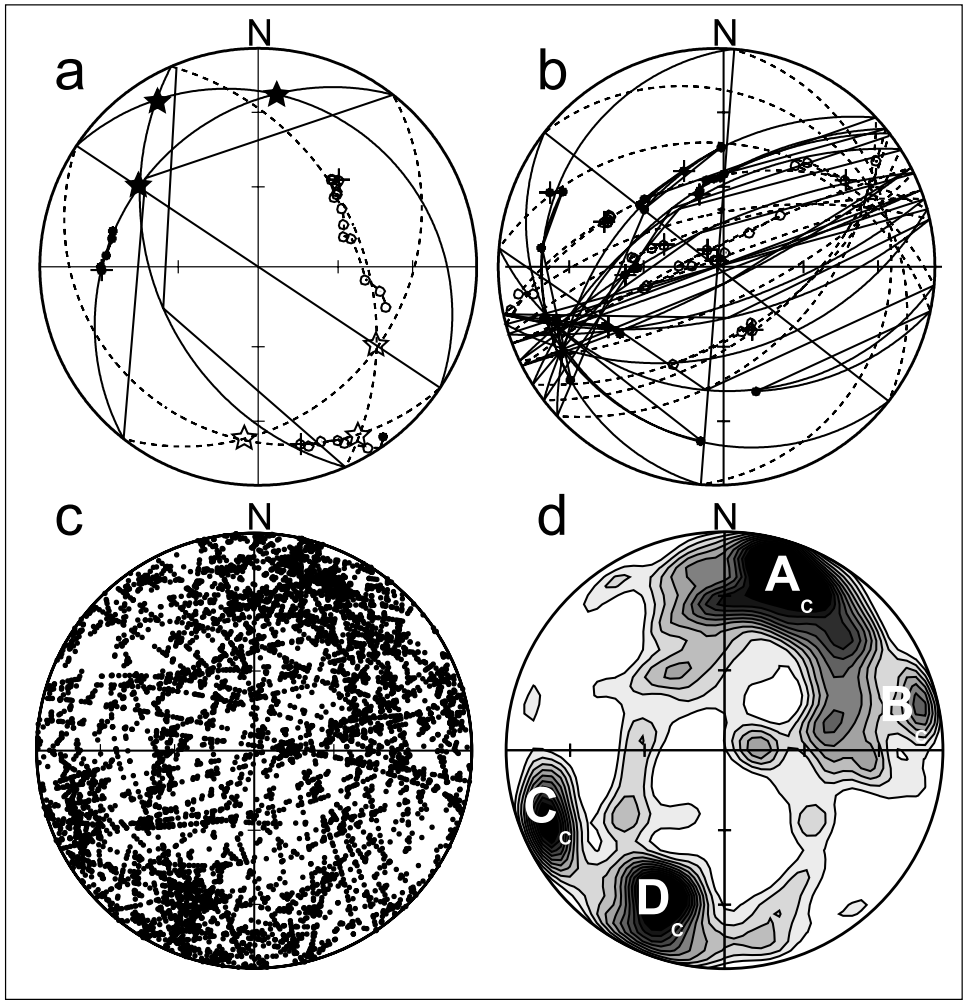
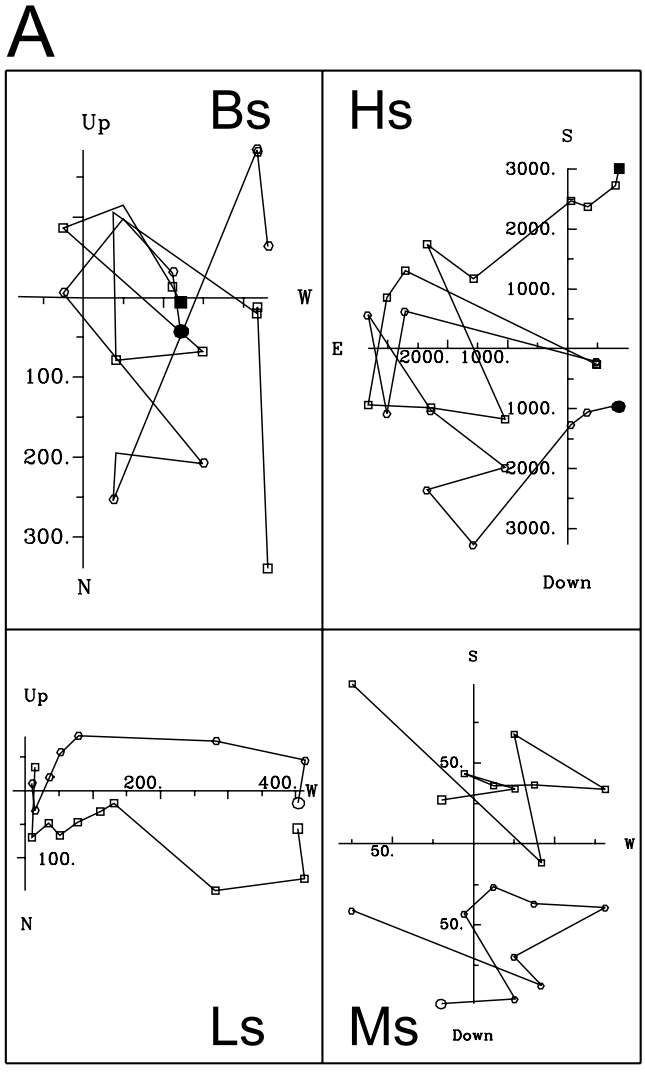


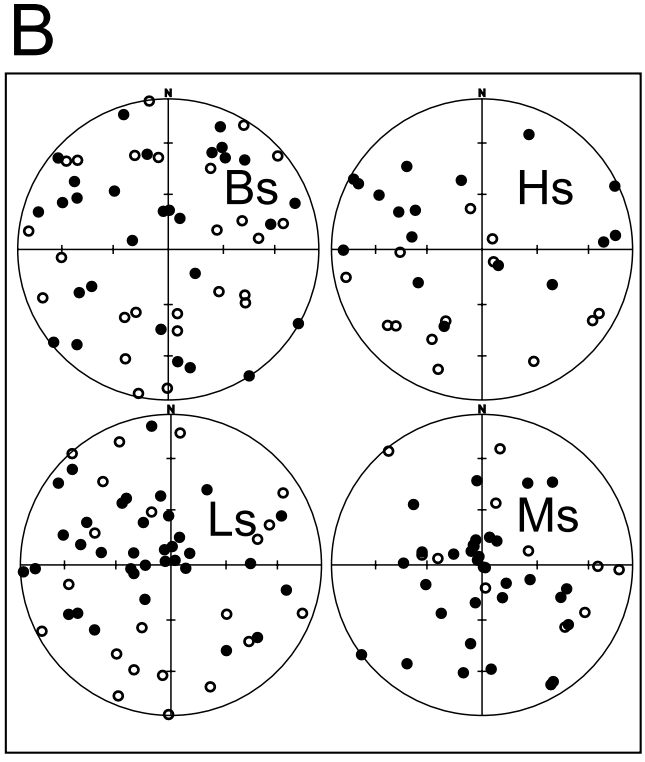
Figure 9. Density distribution diagram for great-circle crossing points, the number of total crossing points = 7711. See text for explanations of the a, b, c, and d stereonets. In Figure 9d the concentration A_c corresponds to ChRM direction for the NSF; the B_c, C_c, and D_c concentrations show other components, probably of Silurian/Devonian age. Schmidt projection (lower hemisphere).

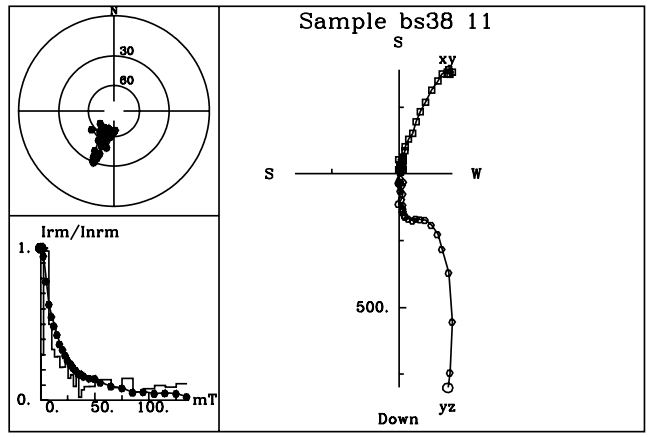
carriers, whereas the younger Balka Quartzite and overlying sandstones on Bornholm and in the Southern Sweden are almost barren of magnetic minerals. In Figure 3 a cartoon of a 350 km long S-N profile of the basement and the sedimentary pile at the end of the Lower Cambrian is shown. Bornholm is situated on the shallow-water shelf of the Upper-Vendian to Cambro-Ordovician Iapetus Ocean/ Tornquist Sea [*Poprawa et al.*, 1999], and the profile illustrates how a regional on-lap progression in deposition of the Cambro-Ordovician sediments was influenced by the basement topography, which may explain the peculiar magnetic feature of these sediments. Early Cambrian erosion of exposed polymetamorphic rocks known in Southern Sweden (Figure 3) some 100–150 km N to NNW of Nekso could supply Ti-hematites and magnetite for the deposition of the NSF, as suggested by *Jensen* [1977] and *Stumpfl* [1958]. Particularly suspected source rocks for the Ti-hematites are the syenites exposed in Southern Scania (see Figure 3). After deposition of the NSF, however, this source was not further exposed due to the deepening of the basin, and therefore it was not available for erosion, the source area for the magnetic grains being covered from the time of deposition of the younger Balka Quartzite and onward.

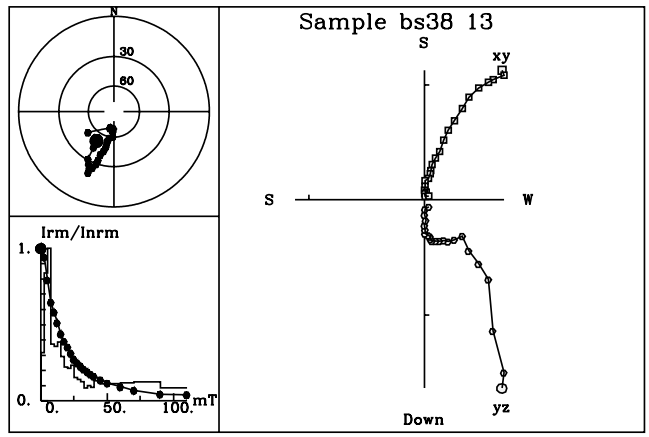
[38] The uniqueness of the paleomagnetic record in the NSF therefore can be explained in terms of the evolution of the early Cambrian sedimentary basin of Bornholm followed by the diagenetic processes that affected the sediments deposited. The source area for the Bornholm basin was gradually submerged and covered by younger formations, which prevented further supply of detrital magnetic grains. This may be inferred from the deterioration of feldspar and martite grains in the upper section of the NSF, combined with a change of environment of deposition from a fluvial and aeolian environment to the shallow marine environment, which triggered accumulation of the Hardeberga (Balka) sandstone [*Jensen*, 1977]. Stimulating a more reductive environment, organic material in the marine sediment could be another factor that influenced the magnetic mineral composition. In summary, coincidence of changes in depositional and source area environments resulted in a new assemblage of magnetic minerals, which particularly was devoid of Ti-poor hematite and Ti-magnetite, the main

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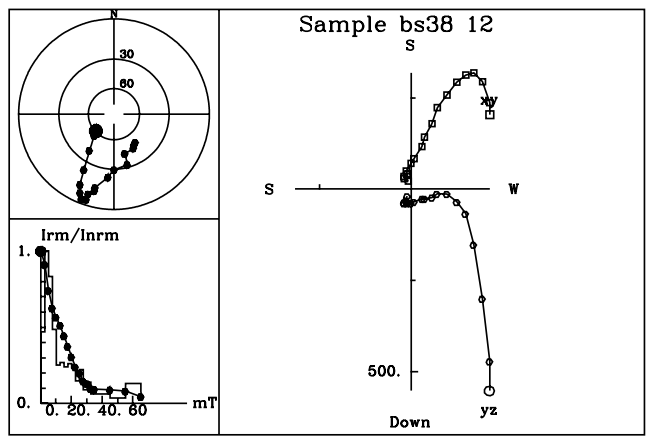


Figure 11. Typical examples of alternating field demagnetization of the Lower Ordovician limestones. The characteristic high coercivity component with the steep inclination is defined at the last stages of demagnetization. (See symbol explanations in Figure 7).

Figure 10. (opposite) (a) Typical response of the Cambrian clastics overlying the NSF to thermal demagnetization; note the zig-zag demagnetization trends in the Zijderveld diagrams. Examples from Balka (Bs), Hardeberga (Hs), Lingula (Ls), and Mickwitzia (Ms) sandstones are shown. (b) Collective sterograms depict the in situ, near-random distribution of the PCA components. (See symbol explanations in Figure 7.)

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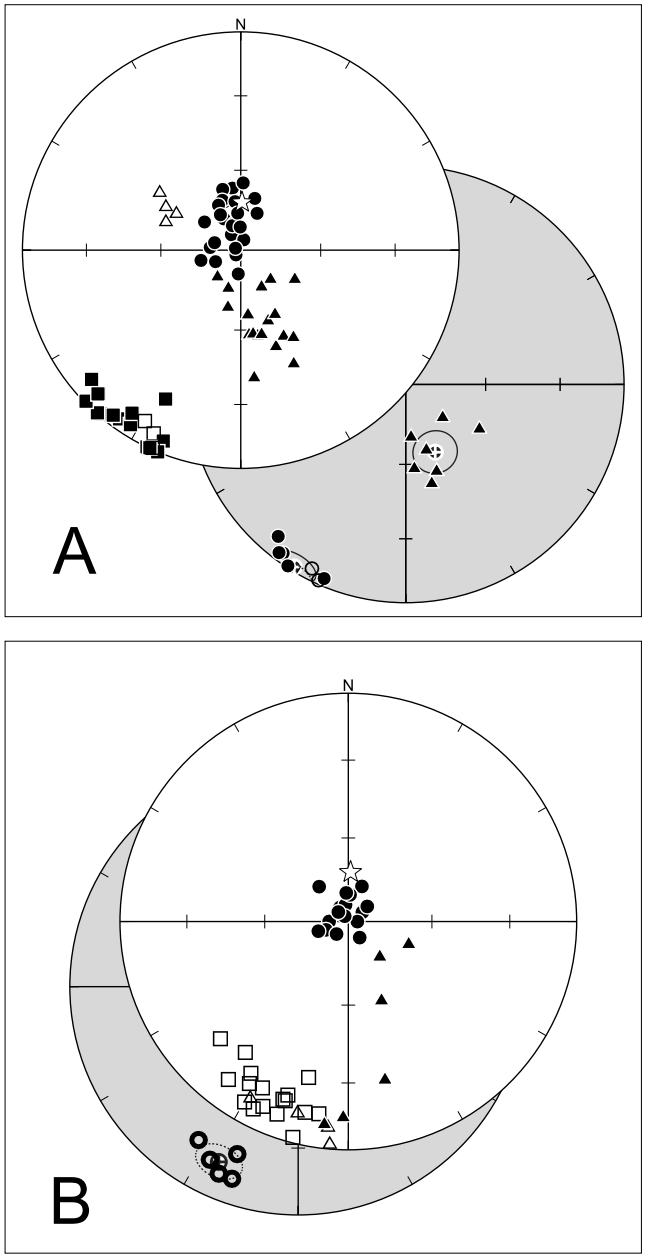


Figure 12. A.F demagnetization of the Ordovician limestones of Bornholm. (a) White stereonet is the distribution of NRM components identified in the Komstad Kalk Formation at Limensgade. Gray stereonet is the sample mean direction (crossed dots) with associated oval of confidence at the 95% probability level. (b). White stereonet is the distribution of the NRM components identified in the Komstad Kalk Formation at Skelbro. Gray stereonet is the sample mean direction (crossed dots) with associated oval of confidence at the 95% probability level. Dots/squares/triangles denote low/intermediate/high coercivity components, respectively. Open/full squares are upward/downward pointing vectors. The star indicates the direction of the present-day geomagnetic field on Bornholm. Schmidt projection.

carrier of the characteristic magnetization of the NSF. Later on, the syn-Caledonian diagenetic events preserved rather than changed the genuine magnetization by closing the porosity and making later fluid migrations ineffective [e.g., *Møller and Friis*, 1999].

[39] The most important argument against the Cambrian age of the characteristic remanence in the NSF is its similarity to the Permian reversed polarity direction for Baltica. This is not a sufficiently compelling argument, however, if a reconstruction of the remagnetization mechanism is attempted. Weathering processes, if affecting the NSF in Permian time, should result in a secondary CRM. If we accept that the observed characteristic magnetization is solely of Permian age, we should also explain what has happened to the original Cambrian magnetization that was undoubtedly present in the Nekso sandstones, given the abundance and variety of magnetic carriers, both of detrital and early-diagenetic origin [*Jensen*, 1977; *Prasad and Sharma*, 1978; *Clemmensen and Dam*, 1993]. These detrital and diagenetic magnetic records should have survived because hematite is chemically very stable and could not be remagnetized by weathering processes alone.

[40] Furthermore, since both isotopic and vitrinitic studies speak in favor of an only mild, not exceeding 200°C, thermal history of the NSF [*Buchardt and Nielsen*, 1985; *Jensenius*, 1987; *Moczydłowska and Vidal*, 1992; *Buchardt et al.*, 1997] for a relatively short time, a thermoviscous resetting of this genuine magnetization can be excluded, at least for the hematite grains of $T_{ub} > 425$°C [cf. *Pullaiach et al.*, 1975]. Therefore even if Permian weathering left a secondary remanence behind (in fact we do not think it did), both the Permian and the Cambrian magnetizations should coexist in the NSF. Because of the lack of other than the observed groups of paleomagnetic directions, the similarity of both Permian and Cambrian remanence directions in the NSF case seems to be an obvious implication.

[41] We cannot answer the question as to how long after deposition the pigment did develop and what is the relative contribution between DRM and CRM. However, taking into account (1) the lack of Permian-like directions in the Lower Cambrian rocks of Southern Sweden and Bornholm (except for the NSF), (2) the presence of detrital magnetic grains and evidence for their relatively fast oxidation, (3) the ambient temperature was not higher than 200°C, precluding a thermal resetting of the genuine Cambrian magnetization residing in the Ti-poor hematite, and (4) the Silurian/Devonian age of the final cementation and pore closure, resulting in high impermeability of the rocks in question for meteoric fluids later on, we may conclude that the characteristic magnetizations residing in the NSF are most probably of primary (detrital) to near-primary (early diagenetic) origin. Taken into account arguments listed above, we are simply unable to envisage a process of acquisition of the late Carboniferous/early Permian magnetization that would totally mask a primary magnetization of the NSF. Even if the Cambrian magnetization and the Permian overprint coexist in the NSF, they must be similar, implicating a subequatorial position of Baltica at the time of deposition of the NSF. Notably, the 583 Ma old Fen complex of S. Norway also shows a characteristic direction similar to the Permian direction but without evidence of being of a secondary origin [*Meert et al.*, 1998]. Finally, our NSF

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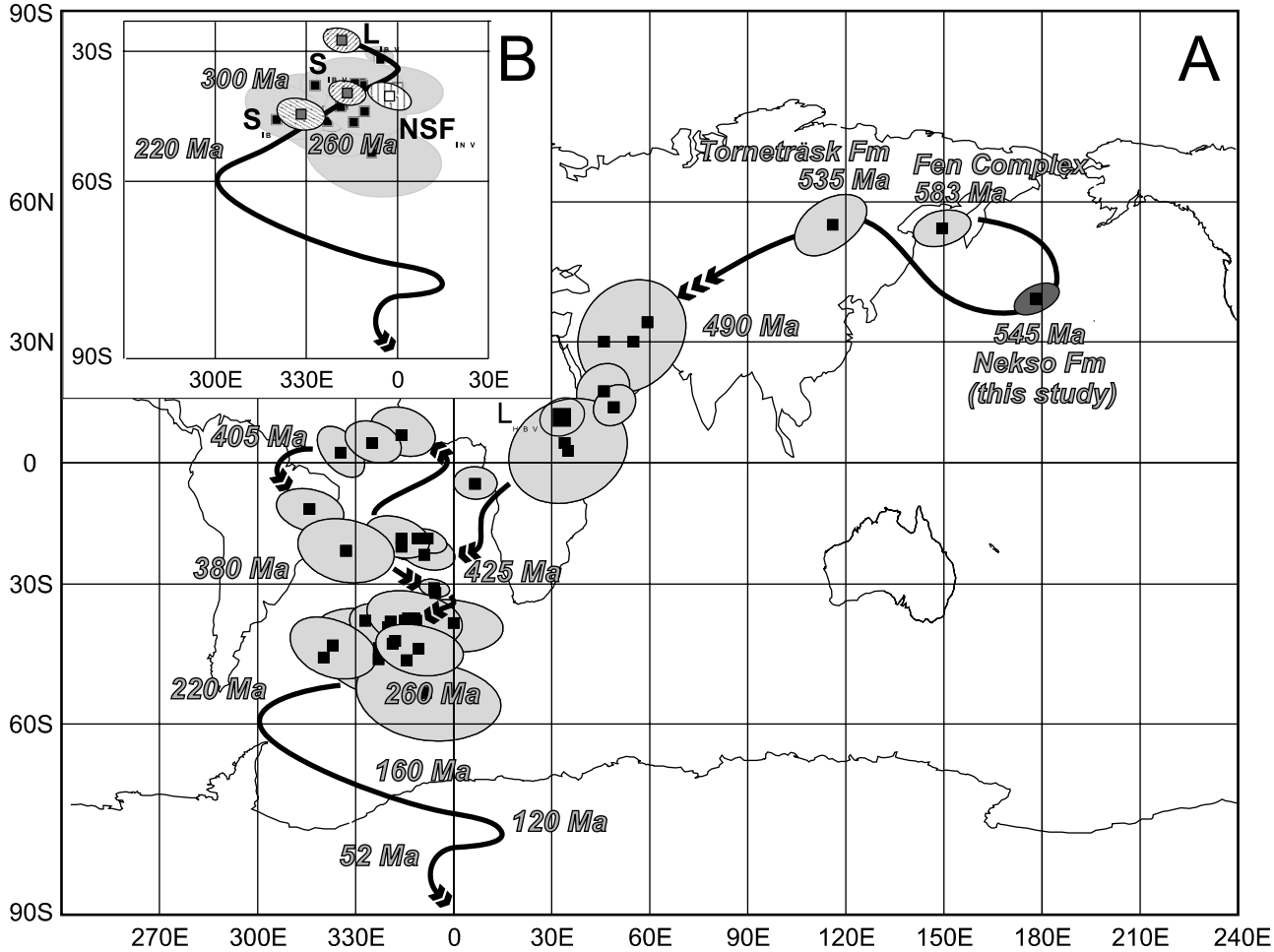


Figure 13. (a) Southern APWP for Baltica marked with solid double arrows. Paleomagnetic poles for the Paleozoic are shown with their ovals of 95% confidence. Time limits for geological periods are given in Ma. The oval of confidence for the NSF paleopole is densely hatched. The paleopole of the Komstad Kalk Formation is marked as a black square. (b) Permo-Carboniferous sector of APWP for Baltica. Paleopoles calculated for LIBV (Limensgade) and SIBV (Skelbro) secondary components are shown with hatchured ovals, and contrasted with paleopole from Skelbro obtained for IB component by *Torsvik and Rehnström* [2002]. NSF paleopole, shown in alternative interpretation with respect to Figure 13a, is show to illustrate its relation to Permo-Carboniferous paleopoles for Batica. Cylindrical projection.

paleopole is almost identical with a pole calculated for the ca. 1630 Ma old quartz porphyry dykes of Baltica [cf. *Buchan et al.*, 2000], suggesting that the Permian-like palaeolatitude for Baltica and its orientation with respect to the Earth's poles might have been a case also earlier in geological history.

[42] Taking into account reported time lags for acquisition of CRM in the red beds [see, e.g., *Butler* [1992] and *Dunlop and Özdemir* [1997] for more comprehensive discussions and literature) we may suggest the time delay of the remanence acquirement after the detrital deposition of a NSF bed to be of the order of 10^4–10^5 y. Low T_{ub} components may be of a polygenetic nature, involving either a diagenetic magnetization stemming from late-diagenetic hematite pigments or a composed nature of these vectors, or both.

[43] Consequently, the proximity of the Nekso paleopole to the Permian poles of Baltica may be considered casual, resulting from a Cambrian orientation of Baltica similar to the much later Permian one. Thus the obtained paleopole for the NSF should be considered representative for the geomagnetic field and for Baltica in the earliest Cambrian (Figure 13), the question about polarity of this paleopole remaining open. Following the principles of a minimum plate velocity and of the shortest distance between paleopoles for the APWP, we propose a normal polarity and a position on the southern hemisphere for Baltica at that time [cf. *Torsvik and Rehnström*, 2001]. Such a paleogeographic position and orientation of Baltica increase, nonetheless, the drift rate for this paleocontinent in the early Cambrian-Lower Ordovician time span and makes the APWP trend more complicated than envisaged by other authors.

5.2. Ordovician Results

[44] None of the NRM components identified in this study can be considered primary. The characteristic HBV component obtained for the Arenigian limestones (~480 Ma) of the Komstad Kalk Formation in Limensgade yielded a

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paleopole position similar to the mid/late Ordovician (~460 Ma) poles of Baltica and hence should be considered secondary (Figure 13). Interestingly, it testifies the presence of an earlier remagnetization event on Bornholm than the one reported by *Torsvik and Rehnström* [2003] for Scania. In turn, HBV components in Skelbro (this study) lie along a great circle between late Paleozoic and early Ordovician directions for Baltica, so a mean direction for this population has not been calculated.

[45] Both in Limensgade and Skelbro the only remanence carrier is magnetite, the origin of which is uncertain. *Torsvik and Rehnström* [2003] considered magnetite of the Skelbro limestone a prime remanence carrier, but they did not explicitly exclude a formation of a secondary magnetite due to authigenesis or diagentic processes. Although primary nature of the magnetite is possible, we are more in favor of its diagenetic origin even though the remanence in Skelbro is similar to the early Ordovician directions of Baltica [see *Torsvik and Rehnström*, 2003]. For instance, the HBV component in the Limensgade limestone is of secondary origin, although not very much younger than the age of the host rocks. Presence of a normal polarity (one hand sample only) in these rocks (see Figure 12) may be tentatively explained in terms of repeated expulsions of reductive fluids in the late Ordovician, a time interval between the events being long enough for a polarity change. According to *Møller and Friis* [1999], the final compaction during late Silurian/early Devonian times was preceded by a hydrocarbon migration, inclusions of which can be identified in the NSF and which may also be found in the Ordovician limestones of Bornholm. Combined with observations by *Torsvik and Rehnström* [2003], one may conclude that reductive brines were periodically migrating in the Scania-Bornholm sector of Baltica during the Ordovician.

[46] The acquisition time for the IBV components in Limensgade and Skelbro can be dated by a comparison of the related paleopoles L_{IBV} and S_{IBV} with the APWP for Baltica. The acquisition times are found to be of late Devonian/early Carboniferous and late Carboniferous ages, respectively (Figure 13, inlet B). The former remagnetization event has not been reported in the literature so far. The latter age estimation is older than that suggested by *Torsvik and Rehnström* [2003], since our S_{IBV} paleopole falls at an older sector of the APWP for Baltica than the paleopole given by *Torsvik and Rehnström* [2003] (see Figure 13). The difference in orientations between the IBV component (this study) and the IB component [*Torsvik and Rehnström*, 2003] may stem from different demagnetization methods applied by the latter authors (thermal cleaning) and by us (A.F. demagnetization); overlapping blocking temperature spectra are not necessarily mirrored in the coercivity distribiutions. Given that the IBV components in this study show Fisherian distribution, while a population of IB components in the work of *Torsvik and Rehnström* [2003] display linear distribution characteristic for overlapping blocking temperatures, we favor our characteristic mean as representative for the ambient geomagnetic field during the remanence acquisition. Conversely, but for the same reason, the ''hard'' components in Skelbro are much better defined in the study by *Torsvik and Rehnström* [2003] than in our case. Regardless of the option, the presence of the late Carboniferous/early Permian NRM overprint in the Skelbro limestones seems to be real.

[47] It must be stressed, however, that a mechanism that led to the origin of this overprint is not adequate to elucidate the origin of the characteristic magnetization of the NSF. Firstly, it features a unique, much localized occurrence, spatially constrained to the vicinity of fault systems throwing down the basement towards the southerly situated TTZ. This suggests causality between tectonics and the overprint. According to *Liboriussen et al.* [1987], the Fennoscandian Border Zone became tectonically (re)activated in Permo-Carboniferous time, inducing a thermal anomaly in the zones of wrench deformations [*Sørensen*, 1987]. A local temperature rise, even up to 200°C [see *Hansen*, 1995] and/or a localized pressure solution due to tectonic stress could partially reset a remanence residing in the magnetite-bearing Ordovician limestones in Skelbro, but we note that the secondary magnetization of the same limestones in Limensgade differ significantly, although the two localities are only two kilometers apart. This may suggest a diachronic development of the fault system on Bornholm.

[48] Second, there is no hematite in the limestones of Skelbro. Its presence would suggest oxidation due to weathering, potentially a causal process for acquisition of the characteristic component in NSF. This is in line with results of studies on stable isotopes composition by *Buchardt and Nielsen* [1985], who ascertained both negligible influence of the recent weathering processes on the limestones in Skelbro and a total lithification of these limestones during syn-Caledonian burial and diagenesis. In turn, occurrence of magnetite carrying a late Paleozoic magnetization speaks in favor of reductive, not oxidizing conditions. This fact suggests the plausible involvement of reductive fluids before the final diagenesis and pore closure (i.e., most probably late Silurian)[cf. *Buchardt and Nielsen*, 1985; see also *Jensenius*, 1987], which provided the iron necessary for development of magnetite. Magnetite grains could be magnetically reset later on due to tectonic activity in Variscan times. Another possibility is a local reheating due to a Permian dyke(s) intrusion, since at least one dyke of this age has been identified in the area, although in the northern part of the island [*Abrahamsen and Lewandowski*, 1992]. Although not encountered in the southern part of Bornholm, some Permian dyke(s) may exist yet undiscovered off the southern shore.

6. Paleogeographic Implication

[49] The subhorizontal paleomagnetic inclinations indicate an equatorial position of Bornholm (and hence of Baltica) during the time of the remanence acquisition (paleolatitude about 8°N or S). Following a minimum plate velocity principle and the shortest distance between paleopoles for the APWP we propose a normal polarity and a southern hemisphere position for Baltica. Such a paleogeographic position and orientation of Baltica do increase the rotation rate for this paleocontinent in the early Cambrian-Lower Ordovician time span and also make its movement more complicated [cf. *Smethurst et al.*, 1997].

[50] The paleolatitude obtained is compatible with that expected for the arid [*Clemmensen and Dam*, 1993] and rather warm climatic conditions, under which red-beds are

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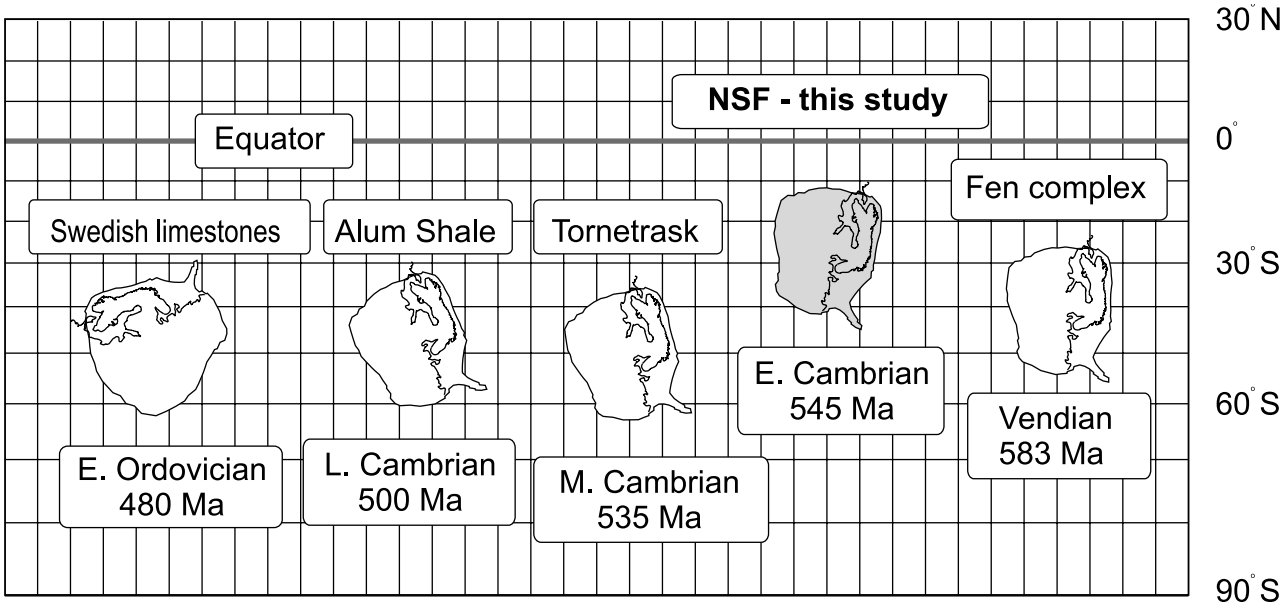


Figure 14. Paleogeographic reconstructions of Baltica according to available paleopoles for the late Neoproterozoic-early Ordovician time span. See text for detailed discussion.

supposed to be deposited [e.g., *Witzke*, 1990; *Turner*, 1980; *Pittman*, 1979]. However, due to the grain size (sand and silty sand fractions) we cannot exclude a contribution of an inclination error (up to say 15°) in the case of the NSF, if the characteristic remanence is of detrital origin [cf. *Tauxe and Kent*, 1984]. On the other hand, we may expect that the high-energy environment of deposition could partially cancel an effect of gravitational torque exerted on the magnetic grains. Also, a Silurian/Devonian compaction could contribute further to shallowing of the remanence. The resulted inclination and compaction errors may, however, only be roughly estimated (say 10°–15°), and this would increase the numerical paleolatitude of 8° for the early Cambrian position of the Bornholm basin up to 15° N or S (Figure 14).

[51] The paleolatitude determined for the NSF is similar to the paleolatitude of 26° found for the 583 Ma Fen tinguaities [*Meert et al.*, 1998]. The Middle Cambrian cold-water limestones (the Exulans Kalk and the Andrarum Kalk Fms.) overlaying the Læseaa Fm. may be interpreted as a herald of the proceeding movement of Bornholm from low (10–15°S) to moderate southern paleolatitudes (35°S) in early Ordovician time. This southward movement of Bornholm was associated with the ca. 90° anticlockwise rotation of Baltica, the Euler pole for this rotation being situated near present-day Novaya Zemlya (cf. Figure 14). We suggest that at the turn of the Precambrian and the Cambrian Baltica was placed in a 180° inverted position as compared to Permian time and it was situated at the southern hemisphere, compatible with a normal polarity for the ChRM of the NSF. An alternative option, with Baltica located at the Northern Hemisphere but in the "Permian" orientation (reversed polarity option for the NSF), would require a similar polarity interpretation for the Torneträsk and Alum Shale results (now being normal) [*Torsvik and Rehnström*, 2001], in consequence leading to exceptionally high plate velocities (ca. 35 cm/year) for Baltica between 500 and 480 Ma. The question of likely plate-tectonic "speed limits" has been addressed by *Meert et al.* [1993]. Finally, the tropical to subtropical paleolatitude suggested compares much better with the position of Baltica as shown by *McKerrow et al.* [1992] than with the configuration presented by *Dalziel* [1992, 1997].

7. Conclusions

[52] The Cambrian sediments at the SW margin of Baltica display strong differences in the paleomagnetic records, arising from their various diagenetic history and tectonic evolution. The fairly chaotic, soft and badly preserved magnetizations of the Balka, the Hardeberga, the Mickwitzia, and the Lingulid sandstones of Bornholm and Southern Sweden contrast with the well-grouped and stable characteristic magnetization of the Nekso Sandstone Formation (NSF).

[53] The regional geological context, including the inferred diagenetic evolution of the Lower Cambrian sediments, points to a syndepositional/early diagenetic origin of the characteristic remanence of the NSF, therefore applicable for plate tectonic interpretations. The inclination of the stable remanence points to a paleolatitude of ca. 15°N or S for Bornholm during the time of the ChRM acquisition. This characteristic component, though similar to the Permian directions of Baltica, is primary or near primary in origin, indicating that the paleogeographic position of Batica in the Cambrian time was similar to that in the Permian. Due to the ambiguity with the polarity being normal or reverse, the question whether the continent should be oriented with respect to the southern or northern paleopole remains open.

[54] The Arenigian limestones of the Komstad Kalk Formation display an NRM with three-components, the hardest one featuring a mixed polarity. The calculated paleopole lies at the slightly younger sector of the APWP for Baltica, suggestive of a secondary, yet Ordovician,

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overprint. Two other shallow and reversed polarity components yielded paleopoles similar to the late Carboniferous-early Permian poles of Baltica, pointing to another remagnetization episode that affected these rocks.

Appendix A

[55] After the first version of the manuscript had been submitted, we got acquainted with an abstract of a paper by *Popov et al.* [2002], published in the *Journal of Geophysical Research, 107*(B11), 2315, 200XXX, 2002, arguing for the subequatorial position of Baltica in the late Vendian (555 Ma) time.

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